Radiative Turbulent Mixing Layers in Astrophysics

Literature Review and a Practical Framework for Computing Emission Spectra

Abstract

Radiative turbulent mixing layers (RTMLs) occur where shearing hot and cold gas mix, pass through the efficient-cooling temperature range, and radiate the enthalpy supplied by the hot phase. This note separates two related but distinct calculations: a bolometric cooling loss and an observable spectrum. It reviews the physical controls on RTMLs and presents an implementable post-processing workflow: obtain the time-dependent thermodynamic history, solve non-equilibrium ionization (NEI) where required, evaluate line and continuum emissivities from an atomic database, integrate them over volume or ray, and finally apply transfer and instrumental response. The central practical result is that a cooling table alone cannot predict a spectrum: the ion fractions and level populations must be specified. In rapidly mixed or rapidly cooled gas, the ionization/recombination time can exceed the cooling or mixing time, making CIE spectra a useful baseline but not the default prediction.

Contents

1. Scope and physical picture

Consider cold gas ($T_c \sim 10^4$ K) moving relative to a hot phase ($T_h \gtrsim 10^6$ K). Kelvin–Helmholtz-driven turbulence mixes material across the interface. At nearly common pressure the mixed gas passes through roughly

$$T_{\rm mix} \sim (T_c T_h)^{1/2}, \tag{1}$$

often near 10^5 K, where metal-line cooling is efficient. In a steady radiative layer, cooling is powered principally by the enthalpy flux of entrained hot gas, with turbulent dissipation potentially contributing. The global energy statement is more robust than any particular geometric model:

$$L_{\rm rad} = \int_V n_{\rm e} n_{\rm H} \Lambda(T, x)\, dV \simeq \dot{M}_{\rm in}\, h_h + \dot{E}_{\rm diss} - \dot{E}_{\rm stored}. \tag{2}$$

Here Λ is the *net* cooling coefficient only when the assumed ionization state and radiation field match the gas. Equation (2) checks the energy budget; it does not determine how $L_{\rm rad}$ is distributed over wavelength.
The relevant competition is captured by

$$\mathrm{Da} \equiv \frac{t_{\rm turb}}{t_{\rm cool}}, \qquad t_{\rm cool} = \frac{e_{\rm th}}{n_{\rm e} n_{\rm H} \Lambda_{\rm net}}. \tag{3}$$

For $\mathrm{Da} \gtrsim 1$, cooling is fast compared with outer-scale mixing and the interface fragments into a strongly corrugated multiphase cooling front. Tan, Oh, and Gronke emphasize that the outer eddy turnover time then sets the effective consumption/mixing rate, whereas Fielding et al. find a corrugated cooling surface with an approximately $5/2$ fractal dimension. These are physical guides for convergence tests, not replacements for a spectrum calculation [8, 7].

2. Development of the existing research

2.1 Analytic foundations and the first spectral predictions

Begelman and Fabian proposed that shear-driven turbulent transport mixes hot and cold material into an intermediate phase with a characteristic temperature near $(T_c T_h)^{1/2}$, and that the incoming hot-phase enthalpy supplies the radiative loss [1]. This model established the two ideas that remain central today: the interface is an energy-conversion layer rather than an isolated parcel, and its luminosity is tied to the rate at which hot gas is entrained. Its assumed layer thickness, often written $h \sim v_{\rm turb} t_{\rm cool}$, should now be regarded as an early closure rather than a universal scaling.
Slavin, Shull, and Begelman converted this picture into emission and absorption predictions for Galactic interfaces [2]. Their calculation included time-dependent ionization and predicted C IV, Si IV, N V, O VI, UV semiforbidden lines, and Hα. This work was conceptually important because it treated ionization lag and spectra together. It also anticipated a persistent observational issue: one individual interface generally produces a modest high-ion column, so a long sightline may need to intersect many interfaces.

2.2 The numerical transition: dimensionality, NEI, and magnetic fields

Esquivel et al. performed three-dimensional MHD calculations with equilibrium cooling and demonstrated that field orientation and finite evolution time matter for the morphology and growth of a mixing layer [3]. Kwak and Shelton subsequently coupled hydrodynamics to NEI rate equations [4]. They found that mixing can transport ions into gas whose temperature differs from the CIE formation temperature, while cooling can outpace recombination. Their simulated C IV/N V/O VI ratios were broadly compatible with Galactic-halo diagnostics, but varied strongly in space and time. Thus a single static "mixing temperature" is insufficient for either ion columns or line emission.

Ji, Oh, and Masterson carried out three-dimensional HD and MHD simulations including NEI and photoionization [6]. Their results revised several early analytic expectations:

- characteristic turbulent and inflow velocities were substantially below the imposed shear velocity and only weakly dependent on it;
- the layer width followed approximately $h \propto t_{\rm cool}^{1/2}$ in their parameter regime, rather than $h \propto t_{\rm cool}$;
- ion columns retained density and metallicity dependence; and
- magnetic amplification and tension could strongly suppress mixing and reduce high-ion columns.

NEI increased the abundances and columns of high ions by factors of a few in relevant cases. Photoionization can also over-ionize the gas and modify its cooling efficiency, although Ji et al. found this secondary for the CGM conditions they explored.

2.3 Fractal fronts, combustion theory, and two cooling regimes

Fielding et al. showed that most cooling occurs in a thin, corrugated sheet whose measured area is consistent with a fractal dimension $D \simeq 5/2$ [7]. Their model connects the increase of cooling area to a hot-gas inflow scaling close to $t_{\rm cool}^{-1/4}$ in the strong-cooling regime. The result explains why a planar interface area can severely underestimate the total emission, but the fitted fractal dimension and cutoff scale should not be treated as universal constants.

Tan, Oh, and Gronke organized the problem using turbulent-combustion theory [8]. In terms of the outer-scale turbulent velocity u' and size L, their simulations distinguish

$$Q,\ v_{\rm in} \propto u'^{1/2} (L/t_{\rm cool})^{1/2}, \qquad \mathrm{Da} < 1 \quad \text{(weak cooling)}, \tag{4}$$

$$Q,\ v_{\rm in} \propto u'^{3/4} (L/t_{\rm cool})^{1/4}, \qquad \mathrm{Da} > 1 \quad \text{(strong cooling)}, \tag{5}$$

where $Q = L_{\rm rad}/A_{\rm projected}$ is the bolometric surface brightness. The apparently conflicting $t_{\rm cool}^{-1/2}$ and $t_{\rm cool}^{-1/4}$ results in the literature can therefore describe different Da regimes. These relations predict a bolometric energy flux, not a wavelength-dependent spectrum.

2.4 Local front structure and observable line predictions

Tan and Oh made a further distinction that is especially important for the present purpose [9]. Global mass entrainment and bolometric cooling can converge when the outer turbulent scale is resolved, even if the microscopic interface is not. In contrast, the temperature PDF, ion columns, and line ratios depend on the local conductive–cooling front. A steady one-dimensional front obeys

$$\frac{d}{dx}\left(\kappa \frac{dT}{dx}\right) = j_x c_p \frac{dT}{dx} + \rho \mathcal{L}(T), \tag{6}$$

where $j_x = \rho v_x$ is the mass flux and $\rho\mathcal{L} = n^2\Lambda - n\Gamma$. The local temperature scale includes the Field length

$$\lambda_{\rm F} = \left(\frac{\kappa T}{n^2\Lambda}\right)^{1/2}. \tag{7}$$

Their 1-D front model reproduced temperature distributions, column densities, and line ratios from 3-D hydrodynamic calculations. It also showed why the temperature dependence of κ matters: Spitzer $\kappa \propto T^{5/2}$, constant conductivity, turbulent diffusion, and grid-scale numerical diffusion weight the emitting temperatures differently. This work supplies a practical subgrid model, but its published spectral tests mostly assumed CIE; an RTML spectrum should combine this local-front treatment with NEI whenever the time-scale test in Section 3.2 fails.

2.5 High-Mach-number and magnetized extensions

Yang and Ji extended plane-parallel RTML calculations to supersonic shear [10]. Below $\mathcal{M} \sim 1$, their bolometric surface brightness and intermediate-ion columns rose roughly as $\mathcal{M}^{1/2}$. Above unity they saturated, because the velocity within the cooling/mixing zone was limited by the local sound speed. Supersonic layers developed a two-zone structure: a nearly Mach-independent cooling/mixing zone and a broader turbulent zone. Turbulent dissipation, rather than hot-phase enthalpy alone, became the dominant power source and cold gas tended toward evaporation rather than condensation.

Recent MHD studies sharpen the magnetic-field caveat. Zhao and Bai found that even initially weak fields can be amplified enough to lower the surface brightness and hot-gas inflow, through both magnetic pressure and suppression of turbulent mixing [11]. Das and Gronke likewise found strong suppression in idealized plane-parallel RTMLs, while driven turbulent boxes showed smaller changes in net cold-gas growth because magnetic fields also altered global morphology and contact area [12]. Therefore an HD scaling cannot simply be multiplied by a universal magnetic correction: the Alfvén Mach number, initial topology, anisotropic conduction, and global geometry must be specified.

2.6 Applications and present observational status

In cloud–wind simulations, Gronke and Oh identified the useful survival and growth condition

$$t_{\rm cool,mix} \lesssim t_{\rm cc}, \tag{8}$$

where $t_{\rm cc}$ is the cloud-crushing time [13]. Cooling of mixed gas then creates new comoving cold material and can allow the cold mass to exceed its initial value. This is an application of RTML physics, although the curved, evolving cloud surface is not identical to a periodic plane-parallel layer.

For cosmological cold streams, Mandelker et al. found that the analogous ratio of the mixed-gas cooling time to the shear-layer growth time controls whether a stream is disrupted or gains mass [14]. Their follow-up radiation model showed that KHI-induced dissipation could reach the luminosity scale of observed Lyα blobs in massive high-redshift haloes [15]. That comparison remains model-dependent because Lyα requires self-shielding, resonant transfer, dust, and the external UV field; bolometric RTML power cannot be equated directly to escaping Lyα luminosity.

Taken together, existing studies support RTMLs as a physically plausible source of intermediate temperature ions and interface radiation. They do not yet establish a unique spectral template. Per-interface columns are often below observed CGM columns, with some models requiring roughly 10^2–10^3 interfaces along a sightline [9, 6]. The number and orientation of interfaces, NEI history, magnetic topology, and radiative transfer are therefore part of the observable model rather than nuisance details.

3. What must be calculated for a spectrum

3.1 From hydrodynamics to an emissivity distribution

For an optically thin cell k, the spectral luminosity is

$$L_\lambda = \sum_k \epsilon_\lambda(T_k, n_{\mathrm{e}k}, f_k, Z, J_\nu) V_k, \tag{9}$$

where f is the vector of ion fractions and J_ν is any external or internally generated radiation field. It is often useful to store a differential emission measure (DEM),

$$\frac{d\mathrm{EM}}{d\log T} = \sum_{k\in d\log T} n_{\mathrm{e}k} n_{\mathrm{H}k} V_k, \qquad \mathrm{EM} = \int n_{\mathrm{e}} n_{\mathrm{H}}\, dV, \tag{10}$$

and convolve it with a CIE contribution function $G_\lambda(T)$:

$$L_\lambda^{\mathrm{CIE}} = \int G_\lambda(T; n_{\mathrm{e}}, Z) \frac{d\mathrm{EM}}{d\log T}\, d\log T. \tag{11}$$

Equation (11) is fast and excellent for a CIE reference model, but it is invalid if f is not the CIE fraction at the local temperature. Do not use a bolometric $\Lambda(T)$ as G_λ; the former has already summed over all radiative channels.
For a resolved transition $u \to l$ the volume emissivity is

$$\epsilon_{ul} = n_u A_{ul} h\nu_{ul}. \tag{12}$$

At low density, statistical equilibrium for the excited levels can usually be solved at fixed ion fraction; collisional excitation, radiative decay, recombination cascades, and (when relevant) photoexcitation enter the rate matrix. The continuum must be added on the same wavelength grid: free–free, free–bound, two-photon, and, where appropriate, dust emission. CHIANTI is well suited to optically thin UV/EUV/optical line and continuum synthesis; AtomDB/APEC is a standard choice for CIE X-ray spectra [16, 17].

3.2 Why NEI is normally necessary in RTMLs

For each element X and charge state q, evolve

$$\frac{df_{X,q}}{dt} = n_{\mathrm{e}} \big[f_{X,q-1} C_{q-1}(T) + f_{X,q+1} \alpha_{q+1}(T) \tag{13}$$

$$- f_{X,q} \{ C_q(T) + \alpha_q(T) \} \big] + \mathcal{P}_{X,q}(J_\nu) + \mathcal{X}_{X,q}, \tag{14}$$

with $\sum_q f_{X,q} = 1$. C, α, $\mathcal{P}$, and $\mathcal{X}$ denote collisional ionization, total recombination, photoionization, and other processes (e.g. charge exchange). Couple this stiff system to the energy equation because the emissivity and cooling rate depend on the evolving fractions. A simple decision rule is to calculate CIE only if both t_{ion} and t_{rec} are short compared with *both* t_{cool} and the local mixing/advection time. Otherwise use NEI.

This is not a small correction in a mixing layer: mixing can be faster than ionization, and cooling can be faster than recombination. Kwak and Shelton and Ji, Oh, and Masterson find enhanced high-ion columns in NEI relative to CIE in relevant RTML regimes [4, 6]. A spectrum intended for O VI, N V, C IV, Si IV, or soft-X-ray diagnostics should therefore report whether it is CIE, NEI, or photoionization+NEI.

4. Recommended calculation workflow

Step 1. Specify the physical experiment. Record T_c, T_h, pressure or densities, relative velocity, metallicity and abundance pattern, magnetic field/conduction assumptions, external radiation field, and the target viewing geometry. State whether the calculation is isochoric, isobaric, or fully hydrodynamic.

Step 2. Obtain a converged flow history. Run a hydrodynamic/MHD calculation with a cooling treatment consistent with the chosen abundances and radiation field, or make a one-dimensional cooling-flow surrogate. Save $(\rho, T, n_{\rm e}, \boldsymbol{v})$ and either Lagrangian tracer histories or sufficiently frequent snapshots. Check that total radiative losses converge with resolution and close the energy budget in Equation (2).

Step 3. Solve the ionization history. Prefer NEI integrated along tracers (or together with the flow). Include H, He, C, N, O, Ne, Mg, Si, S, and Fe at minimum for broad UV–X-ray coverage, plus charge exchange near the cool interface. Use an implicit/BDF solver or another positivity-preserving stiff integrator. As a controlled diagnostic, generate an otherwise identical CIE spectrum and quote the ratio line by line.

Step 4. Evaluate atomic emissivities. For each saved cell/tracer, compute level populations and ϵ_λ using its $T, n_{\rm e}$ and NEI fractions. CHIANTI/ChiantiPy is convenient for optically thin line work; AtomDB/PyAtomDB is convenient for CIE X-rays. For photoionized, optically thick, or radiation-coupled zones, use a spectral-synthesis/transfer code such as CLOUDY, but supply the non-equilibrium state or a sequence of short history steps rather than silently imposing equilibrium.

Step 5. Project and transfer the radiation. The intrinsic surface brightness along a ray is

$$I_\lambda = \frac{1}{4\pi} \int \epsilon_\lambda(s) \exp[-\tau_\lambda(s)]\, ds. \tag{15}$$

For an optically thin forbidden line set $\tau_\lambda = 0$. Treat resonance lines (e.g. Lyα, C IV, O VI), dust UV extinction, and X-ray photoelectric absorption explicitly whenever their columns make them non-negligible. Add thermal and turbulent Doppler shifts before binning; then convolve with the telescope line-spread function and effective area.

Step 6. Validate and publish reproducibly. Compare the integrated model spectrum against the cooling loss, $\int L_\lambda d\lambda \simeq L_{\rm rad}$ after accounting for untracked bands. Test atomic-data version, time cadence, resolution, abundances, CIE versus NEI, and viewing angle. Publish line luminosities, intrinsic and attenuated spectra, and the DEM, rather than only selected line ratios.

5. A minimal, useful surrogate: a steady isobaric cooling flow

When a full 3-D RTML calculation is unavailable, a steady cooling-flow model gives a transparent first estimate. Assume a mass flux per interface area $\dot{m}/A$ and evolve a parcel from its initial mixed temperature under constant pressure. With enthalpy per mass h, solve

$$\frac{dh}{dt} = -\frac{n_{\rm e} n_{\rm H} \Lambda_{\rm net}(T, f)}{\rho}, \quad \frac{df}{dt} = R(T, n_{\rm e}, J_\nu) f, \tag{16}$$

then compute a line luminosity per interface area,

$$\frac{L_{ul}}{A} = \frac{\dot{m}}{A} \int \frac{\epsilon_{ul}}{\rho} \, dt. \tag{17}$$

This has two benefits: it makes the assumed energy/mass flux explicit, and it handles ionization lag. Its limitation is equally explicit: it cannot predict the entrainment rate, corrugated area, or velocity-dependent line profile. Those require a turbulence model or simulation. Use it as a benchmark and parameter study, not as a substitute for resolved RTML dynamics.

6. Diagnostics and observables

The expected output is multi-band. The $\sim 10^{4.5}$–$10^{5.5}$ K material can produce UV high-ion diagnostics (C IV, Si IV, N V, O VI); hotter gas contributes EUV/soft-X-ray lines and continuum; recombined material can emit optical/IR lines if it remains ionized or is externally illuminated. The classical mixing-layer models of Slavin, Shull, and Begelman explicitly connected high ions and UV/optical emission to a turbulent interface [2].

Interpret line ratios jointly with the DEM and absorption columns:

$$N_{X,q} = \int n_X f_{X,q} \, ds. \tag{18}$$

Emission scales approximately as density squared in the optically thin, low-density limit, while absorption scales linearly with density. Combining them constrains filling factor and path length far better than either alone. NEI may make an ion appear at a temperature where its CIE contribution function is small; a "formation temperature" is therefore a model-dependent description, not a direct thermometer in an RTML.

7. Numerical and physical failure modes

- **Cooling–spectrum mismatch:** a tabulated CIE cooling curve is used in the dynamics while post-processing assumes NEI or a different abundance set. Recompute or quantify the inconsistency.
- **Under-resolved numerical mixing:** numerical diffusion can set the DEM and artificially boost cooling. Demonstrate convergence of integrated luminosity, DEM, and selected line luminosities, not merely morphology.
- **Equilibrium by assumption:** CIE can miss the lagging high ions in cooling/mixing gas. Carry NEI fractions, at least in post-processing.
- **Ignoring transfer:** optically thin synthesis is not adequate for strong resonance lines or dusty/high-column sightlines.

- **Comparing unlike quantities:** distinguish intrinsic luminosity, surface brightness, attenuated flux, and detector counts; each has different geometry and units.

8. Literature context

Study	Main method	Reliable takeaway for spectral work
Begelman & Fabian (1990)	Analytic energy balance	Establishes the enthalpy–cooling balance that remains a necessary global check.
Slavin, Shull & Begelman (1993)	Turbulent-interface/NEI spectral models	Direct precedent for predicting high-ion emission and absorption from mixing layers.
Esquivel et al. (2006)	3-D MHD + CIE cooling	Demonstrates time-dependent layer growth and magnetic sensitivity; CIE limits its spectral interpretation.
Kwak & Shelton (2010)	Hydro + NEI	Mixing and cooling histories enhance high ions relative to CIE; time dependence matters.
Ji, Oh & Masterson (2019)	3-D HD/MHD + NEI + radiation	Simple $h \sim v_{\rm turb} t_{\rm cool}$ scalings can fail; NEI changes high-ion columns by factors of a few.
Fielding et al. (2020)	3-D hydro, fractal cooling front	The cooling surface is corrugated; use global energetics and resolution tests rather than planar geometry alone.
Tan, Oh & Gronke (2021)	Turbulent-combustion analogy	Da organizes weak- and strong-cooling regimes and informs convergence tests.
Tan & Oh (2021)	1-D front + 3-D spectral tests	Global entrainment and line observables have different resolution requirements; local conduction controls the temperature PDF.
Yang & Ji (2023)	High-Mach 3-D hydro	Surface brightness and ion columns saturate above $\mathcal{M} \sim 1$; dissipation can drive evaporation.
Zhao & Bai (2023); Das & Gronke (2024)	3-D MHD	Amplified weak fields suppress local mixing, while global morphology can partly compensate in turbulent systems.

9. Conclusions

An RTML radiation spectrum is a state-resolved emissivity calculation, not a direct plot of $\Lambda(T)$. The recommended deliverable is a NEI-based intrinsic spectrum plus transferred spectra for specified sightlines, accompanied by a DEM, ion columns, and an energy-closure test. A CIE/DEM spectrum remains valuable as a fast control model; differences from NEI then become an interpretable prediction of the mixing and cooling history.

References

[1] Begelman, M. C. & Fabian, A. C. 1990, MNRAS, 244, 26P, doi:10.1093/mnras/244.1.26P.

[2] Slavin, J. D., Shull, J. M. & Begelman, M. C. 1993, ApJ, 407, 83, doi:10.1086/172494.

[3] Esquivel, A., Benjamin, R. A., Lazarian, A., Cho, J. & Leitner, S. N. 2006, ApJ, 648, 1043, doi:10.1086/505338.

[4] Kwak, K. & Shelton, R. L. 2010, ApJ, 719, 523, doi:10.1088/0004-637X/719/1/523.

[5] Gnat, O. & Sternberg, A. 2007, ApJS, 168, 213, doi:10.1086/509786.

[6] Ji, S., Oh, S. P. & Masterson, P. 2019, MNRAS, 487, 737, doi:10.1093/mnras/stz1248.

[7] Fielding, D. B., Ostriker, E. C., Bryan, G. L. & Jermyn, A. S. 2020, ApJL, 894, L24, doi:10.3847/2041-8213/ab8d2c.

[8] Tan, B., Oh, S. P. & Gronke, M. 2021, MNRAS, 502, 3179, doi:10.1093/mnras/stab053.

[9] Tan, B. & Oh, S. P. 2021, MNRAS Letters, 508, L37, doi:10.1093/mnrasl/slab100.

[10] Yang, Y. & Ji, S. 2023, MNRAS, 520, 2148, doi:10.1093/mnras/stad264.

[11] Zhao, X. & Bai, X.-N. 2023, MNRAS, 526, 4245, doi:10.1093/mnras/stad3011.

[12] Das, H. K. & Gronke, M. 2024, MNRAS, 527, 991, doi:10.1093/mnras/stad3125.

[13] Gronke, M. & Oh, S. P. 2018, MNRAS Letters, 480, L111, doi:10.1093/mnrasl/sly131.

[14] Mandelker, N., Nagai, D., Aung, H., Dekel, A., Birnboim, Y. & van den Bosch, F. C. 2020, MNRAS, 494, 2641, doi:10.1093/mnras/staa812.

[15] Mandelker, N., van den Bosch, F. C., Nagai, D., Dekel, A., Birnboim, Y. & Aung, H. 2020, MNRAS, 498, 2415, doi:10.1093/mnras/staa2421.

[16] Del Zanna, G., et al. 2021, ApJ, 909, 38, doi:10.3847/1538-4357/abd8ce; CHIANTI database.

[17] Foster, A. R., et al. 2012, ApJ, 756, 128, doi:10.1088/0004-637X/756/2/128; AtomDB.